Boyer–Moore Voting Algorithm

MAJORITY ELEMENT · I

The Original Problem (> n/2)

Suppose we have:

```
[2, 2, 1, 1, 1, 2, 2]
```

The majority element is **2** because it appears 4 times out of 7.

The naive solutions are:

- Brute Force → $O(n^2)$
- Hash Map → $O(n)$, $O(n)$ space

> *Can we do O(1) space? — Yes.*

The Main Idea

Imagine every occurrence of the majority element is a **vote**. Whenever we see a different element, they cancel each other.

2 vs 1

One vote of 2, one vote of 1 — both disappear.

After cancelling all different pairs, the majority element must remain, because it occurs more than all the other elements combined.

```
2   2   1   1   1   2   2
```

Cancel pairs:

`2, 1`	→ *gone*
`2, 1`	→ *gone*
`1, 2`	→ *gone*
`2`	→ **remains**

The winner is still **2**

> *This is the entire intuition.*